

SECTION 3: FORTRAN LOOPS

We would like to repeat a sequence of statements over and over

This repetition can be controlled one of two ways:

- 1) a counter (counting DO Loops)
- 2) a logical expression (WHILE loops)

Lecture 11: Counting DO Loops

General Form:

```
[name] : DO index = istart, iend, increment
```

```
    Statement 1
```

```
    Statement 2
```

```
    ...
```

```
END DO [name]
```

Example to compute e^x by summing series representation:

```

!CODE TO COMPUTE  $e^x=1+x+x^2/2!+x^3/3!+...$ 
PROGRAM e_to_x
IMPLICIT none
REAL :: x, e_x, next_term
INTEGER :: i, num_terms
PRINT*
WRITE(*, '(a)', ADVANCE='NO') 'Enter the value for the exponent x: '
READ(*,*)x
WRITE(*, '(a)', ADVANCE='NO') 'Enter the number of terms to keep in
the series (2 minimum, 50 maximum): '
READ(*,*)num_terms
IF(num_terms > 50)num_terms=50
IF(num_terms < 2)num_terms=2
e_x=1.0
next_term=x
DO i=2,num_terms,1
    e_x = e_x + next_term
    next_term=next_term*(x/REAL(i))
END DO
PRINT*, 'Keeping', num_terms, 'terms', ' e^', x, '=', e_x
PRINT*, 'This compares to the Fortran EXP(X) &
        function result: ', EXP(x)
PRINT*, 'The absolute error is:', ABS(EXP(x)-e_x)
END PROGRAM e_to_x

```

Results for e^x calculations:

```
cmd.exe - Shortcut
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>gfortran e_to_x.f90
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe
Enter the value for the exponent x: 1
Enter the number of terms to keep in the series (2 minimum, 50 maximum): 3
Keeping 3 terms  $e^1 = 2.50000000$ 
This compares to the Fortran EXP(X) function result: 2.71828175
The absolute error is: 0.218281746
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe
Enter the value for the exponent x: 1
Enter the number of terms to keep in the series (2 minimum, 50 maximum): 10
Keeping 10 terms  $e^1 = 2.71828175$ 
This compares to the Fortran EXP(X) function result: 2.71828175
The absolute error is: 0.00000000
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe
Enter the value for the exponent x: 10
Enter the number of terms to keep in the series (2 minimum, 50 maximum): 5
Keeping 5 terms  $e^{10} = 644.333252$ 
This compares to the Fortran EXP(X) function result: 22026.4648
The absolute error is: 21382.1309
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe
Enter the value for the exponent x: 10
Enter the number of terms to keep in the series (2 minimum, 50 maximum): 50
Keeping 50 terms  $e^{10} = 22026.4648$ 
This compares to the Fortran EXP(X) function result: 22026.4648
The absolute error is: 0.00000000
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>gfortran -fdefault-real-8 e_to_x.f90
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe
Enter the value for the exponent x: 10
Enter the number of terms to keep in the series (2 minimum, 50 maximum): 50
Keeping 50 terms  $e^{10} = 22026.465794806714$ 
This compares to the Fortran EXP(X) function result: 22026.465794806718
The absolute error is: 3.6379788070917130E-012
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>
```

Lecture 12: Cycle and Exit statements

Control the operation of Counting Do loops and While loops (next lecture)

CYCLE: if executed within body of loop returns control back to the top of the loop

EXIT: if executed within the body of loop, control goes to first statement after loop

Example:

```
!      PROGRAM TO DEMONSTRATE CYCLE and EXIT STATMENTS  
      PROGRAM cycle_exit  
      IMPLICIT NONE  
      INTEGER :: i  
      DO i=1,20  
          IF(i >5 .AND. i<10)CYCLE  
          IF(i .eq. 15)EXIT  
          PRINT*, 'i=: ',i  
      END DO  
      END PROGRAM cycle_exit
```



```
cmd.exe - Shortcut  
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe  
i=: 1  
i=: 2  
i=: 3  
i=: 4  
i=: 5  
i=: 10  
i=: 11  
i=: 12  
i=: 13  
i=: 14  
G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>
```

Lecture 13: While Loops

General Form:

```
[name] DO
```

```
    IF (logical_expression) EXIT
```

```
    Statement1
```

```
    Statement2
```

```
    ...
```

```
END DO [name]
```

Example guessing random number between 0 and 10:

```

!   program to guess random numbers
PROGRAM guess_number
IMPLICIT NONE
REAL :: r
INTEGER :: number, guess, istat
PRINT*

CALL RANDOM_SEED()
CALL RANDOM_NUMBER(r)
number=NINT(r*10.)

print*,number  !statement for debugging
loop:DO
WRITE(*,'(a) ',ADVANCE='NO') 'Enter an integer between 0 and 10: '
READ(*,*,iostat=istat)guess
    outer_if : IF (istat/=0) THEN !not an integer number
CYCLE loop
    ELSE !valid integer number
        IF(guess <0 .OR. guess>10)THEN
CYCLE loop
        ELSE IF(guess == number)THEN
            PRINT*,'Correct Guess'
EXIT loop
        ELSE
            PRINT*,'Incorrect Guess, Try Again'
        END IF
    END IF outer_if
END DO loop

END PROGRAM guess_number

```

cmd

cmd.exe - Shortcut

G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>a.exe

7

Enter an integer guess between 0 and 10: 4

Incorrect Guess, Try Again

Enter an integer guess between 0 and 10: 8

Incorrect Guess, Try Again

Enter an integer guess between 0 and 10: 4.1

Enter an integer guess between 0 and 10: -1

Enter an integer guess between 0 and 10: 23

Enter an integer guess between 0 and 10: 7

Correct Guess

G:\FORTRAN_COURSE\SECTION3-LOOPS\CODES>